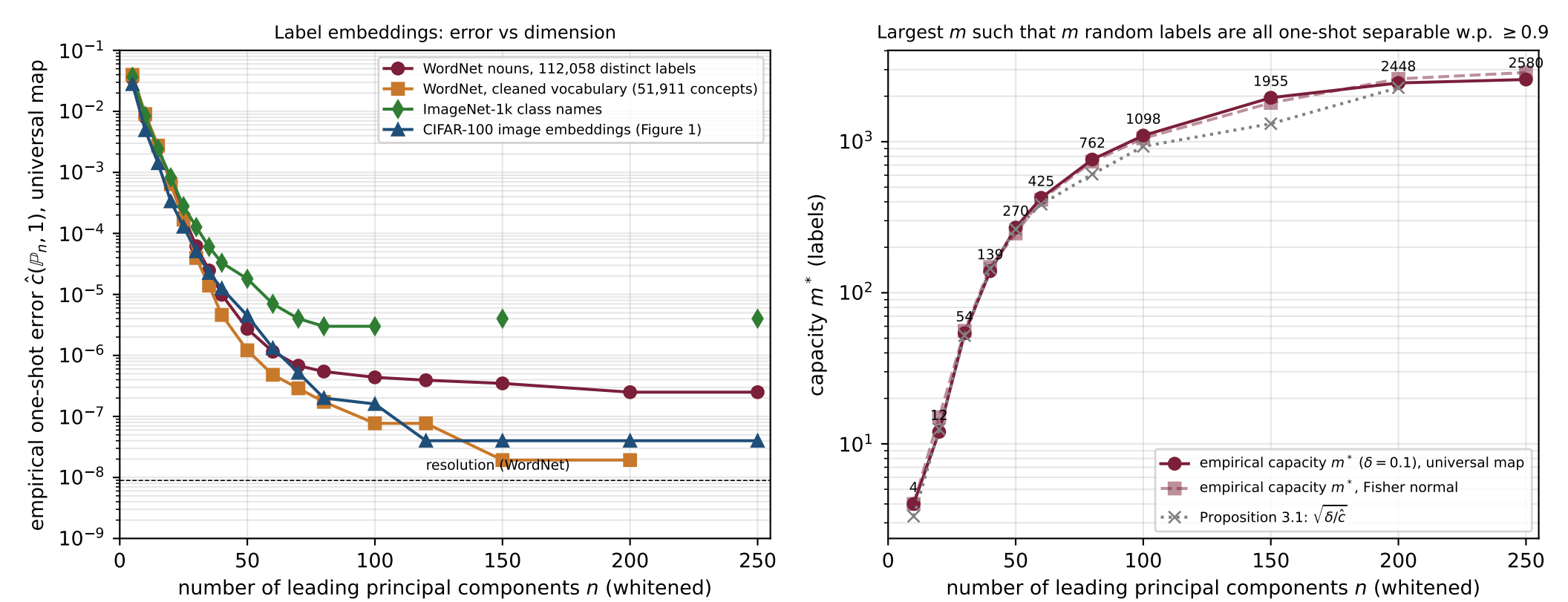
Label embeddings: error vs dimension
empirical one-shot error $\hat{c}(\mathbb{P}_n, 1)$, universal map
number of leading principal components n (whitened)
WordNet nouns, 112,058 distinct labels
WordNet, cleaned vocabulary (51,911 concepts)
ImageNet-1k class names
CIFAR-100 image embeddings (Figure 1)
resolution (WordNet)
Largest m such that m random labels are all one-shot separable w.p. ≥ 0.9
capacity m^* (labels)
number of leading principal components n (whitened)
empirical capacity m^* ($\delta = 0.1$), universal map
empirical capacity m^*, Fisher normal
Proposition 3.1: $\sqrt{\delta/\hat{c}}$
4
12
54
139
270
425
762
1098
1955
2448
2580